Nakuja Project

Solid Propulsion Team

N4 Four-Grain Static Test — Technical Report

Stainless Steel Casing, Mild Steel Bulkhead and Nozzle

Document Version:	1.1 (Post-Execution Report, motor specifications added)
Motor Configuration:	N4, Four-Grain, Stainless Steel (304) Casing
Propellant:	Pure KNSB — no suppressant or additives
Date of Execution:	Friday, 16 October 2026
Prepared by:	Paschal Peters (Data Recorder / Timer)
Outcome:	**SUCCESS** (with one noted anomaly) — full burn achieved; gas leakage at the bulkhead side shortly after ignition

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1 Purpose

This report documents the N4 four-grain static test conducted on Friday, 16 October 2026. The test was a repeat of the four-grain configuration using **pure KNSB grains with no suppressant or additives**, carried out with substantially revised motor hardware. The aluminium bulkhead used in the previous four-grain static test melted during that burn, so the hardware was changed for this test: the casing was changed to **Stainless Steel 304**, the bulkhead was changed from aluminium to **mild steel**, and the nozzle was made of **mild steel**.

The motor burned to completion with a burn time of approximately 4.277 s. The main anomaly was **gas leakage from the bulkhead side shortly after ignition**, suspected to result from differential thermal expansion between the bulkhead and the nozzle combined with failure of the O-ring seal. This report follows the team's updated format: it presents the theoretical configuration (grain criteria and igniter mass), the simulation, the experimental results, an analysis of the leakage anomaly, and the corrective actions to be implemented before the next test.

2 Test Overview

Motor	N4, Four-Grain
Propellant	Pure KNSB (control formulation, no additives)
Experimental Burn Time	≈4.277 s
Simulated Burn Time	4.560 s
Main Anomaly	Gas leakage from the bulkhead side shortly after ignition
Suspected Cause	Differential thermal expansion (bulkhead vs. nozzle) and O-ring failure
Outcome	**Success** — burn completed; leakage anomaly recorded

2.1 Hardware Configuration for This Test

Component	Material	Dimensions and Notes
Casing	Stainless Steel 304 (G304)	765 mm long; 108 mm inner diameter, 114 mm outer diameter, 3 mm wall. Changed for this test.
Bulkhead	Mild Steel	110 mm outer diameter, 52 mm inner diameter, 25 mm bore depth, 50 mm length. Replaced the aluminium bulkhead, which melted in the previous four-grain static test.
Nozzle	Mild Steel	25 mm throat diameter; 30° convergence angle, 15° divergence angle; 52 mm upper length, 93 mm lower length; wall thickness 10 mm at the throat and 15 mm at the port.
Liner	—	2 mm thickness
Bulkhead sealing	O-ring(s)	Material, size, and groove dimensions to be recorded (see Section 8.5).

3 Theoretical Section — Grain Criteria and Configuration

3.1 Grain Details

Four individually prepared pure KNSB grains (no surfactant, suppressant, or other additives) were assembled into the casing for this test. All four grains share the same geometry. Each grain was reported as about 1 kg; individual weighed masses were not recorded (Section 3.3 cross-checks this figure).

Grain	Mass (reported)	Length (mm)	Core Dia. (mm)	Outer Dia. (mm)	Liner (mm)
Grain One	≈1 kg	150	34	104	2
Grain Two	≈1 kg	150	34	104	2
Grain Three	≈1 kg	150	34	104	2
Grain Four	≈1 kg	150	34	104	2
Total (Propellant)	≈4 kg	—	—	—	—

The grain outer diameter plus a 2 mm liner on each side (104 mm + 2 × 2 mm = 108 mm) matches the casing inner diameter of 108 mm.

3.2 Assembly and Configuration Details

Number of Grains	4
Propellant	Pure KNSB, no additives
Inter-grain Spacing	20 mm × 4 (as recorded)
Casing	Stainless Steel 304; 765 mm long; 108 mm ID × 114 mm OD (3 mm wall)
Bulkhead	Mild Steel; 110 mm OD, 52 mm ID, 25 mm bore depth, 50 mm length
Nozzle	Mild Steel; 25 mm throat diameter; 30° convergence and 15° divergence angles; 52 mm upper length, 93 mm lower length; wall thickness 10 mm (throat) and 15 mm (port)
Total Propellant Mass (reported)	≈4 kg
Igniter Mass	*[to be recorded]*
Total Charge Mass (Propellant + Igniter)	*[to be recorded]*

As in the three-grain report, the igniter mass is recorded as part of the theoretical charge configuration, since it contributes to the total initial mass loaded into the motor and is an input to the corresponding OpenMotor simulation. It was not included with the data supplied for this report.

Taking the 20 mm × 4 spacing as four 20 mm gaps, the grain stack occupies 4 × 150 mm + 4 × 20 mm = 680 mm of the 765 mm casing, leaving 85 mm for the end hardware.

3.3 Derived Geometry and Mass Cross-Check

Quantity	Value
Grain web thickness (outer dia. − core dia.)/2	35 mm

Quantity	Value
Grain volume (each)	≈1138 cm^3
Mass per grain at 1750 kg/m^3	≈1.99 kg (≈7.97 kg for four)
Nozzle throat area	≈491 mm^2
Initial port area (one grain core)	≈908 mm^2
Initial port-to-throat area ratio	≈1.85
Nozzle exit diameter (estimate)	≈49.5 mm ($\varepsilon \approx 3.9$) if the listed angles are included angles; ≈75 mm ($\varepsilon \approx 9$) if they are half-angles

Mass cross-check. The reported mass of about 1 kg per grain is roughly half the mass implied by the grain geometry. At 1750 kg/m^3, a 150 mm grain with a 104 mm outer diameter and 34 mm core works out to about 1.99 kg, or about 8 kg for the four grains. The measured impulse supports the geometric figure. Dividing the experimental impulse of 6909 Ns by 4 kg gives a delivered specific impulse of about 176 s, and the simulated 10042 Ns gives about 256 s; both are well above what KNSB can deliver (its ideal specific impulse is roughly 130 s, and the team's single-grain N4 test delivered about 80 s). Using about 8 kg gives roughly 88 s experimental and 128 s simulated, which are plausible. The grains (or the unburned remainder) should be weighed to settle this before any specific impulse is quoted.

Port-to-throat ratio. The initial port-to-throat area ratio of about 1.85 is lower than the 2.32 of the single-grain N4 motor and below the minimum of about 2 that is commonly recommended for KNSB motors. In a four-grain stack the gas from all upstream grains flows through the cores of the downstream grains, so port gas velocity is highest at the aft end, where erosive burning is most likely. This is discussed in Section 5.4.

Nozzle exit. The nozzle exit diameter was not recorded. If 30° and 15° are included angles, the listed lengths give an exit diameter of about 49.5 mm and a convergent inlet diameter of about 53 mm, which matches the 52 mm bulkhead bore. If they are half-angles, the exit is about 75 mm and the inlet about 85 mm. The exit diameter should be measured and recorded.

Dimension check. As recorded, the bulkhead outer diameter (110 mm) is larger than the casing inner diameter (108 mm), so the bulkhead cannot be a plain insertion fit inside the casing bore. This should be confirmed (for example, a recording error, or a design in which part of the bulkhead sits outside the casing), because the clearances and groove dimensions at this joint are central to the leak analysis in Section 6.

3.4 Material Properties Relevant to Sealing

The change of casing, bulkhead, and nozzle materials changes the thermal behaviour of the sealed joints. The table below lists typical handbook values for the materials involved; they are indicative only and should be replaced with the supplier's data for the actual stock used.

Property (typical)	Stainless Steel 304	Mild Steel	Aluminium (6000 series)
Thermal expansion coefficient (10^{-6}/K)	≈17	≈12	≈23

Property (typical)	Stainless Steel 304	Mild Steel	Aluminium (6000 series)
Thermal conductivity (W/m K)	≈16	≈50	≈200
Melting range (°C)	≈1400–1450	≈1450–1520	≈600–650

Two points follow from this table. First, KNSB combustion temperatures are well above the melting range of aluminium, which is consistent with the aluminium bulkhead melting in the previous four-grain test; steel components have a far larger margin. Second, Stainless Steel 304 expands roughly 40–45% more per degree than mild steel, so with a stainless casing and mild steel end hardware the casing tends to grow away from the bulkhead and nozzle as the assembly heats. This matters for any seal that relies on a fixed radial squeeze (see Section 6).

4 Simulation

The N4 four-grain configuration was modelled in OpenMotor prior to testing, using the grain geometry, spacing, and igniter mass described in Section 3. The simulation predicts the following performance:

Parameter	Simulated Value
Peak Thrust	2434.858 N
Time of Peak Thrust	1.710 s
Total Impulse	10041.697 Ns
Burn Time	4.560 s
Average Thrust (impulse / burn time)	≈2202 N

The simulated curve (N4_4grain) rises almost instantly to about 2200 N, climbs gently to a broad maximum near 1.7 s, and then declines slowly. Near the end it steps down in stages — from roughly 1850 N to about 1300 N, then about 1240 N, then a short low tail — as the grains burn out in sequence, ending at 4.56 s.

5 Results

5.1 Experimental Performance

Parameter	Experimental Value
Peak Thrust	4154.460 N
Time of Peak Thrust	0.822 s
Total Impulse	6909.259 Ns

Parameter	Experimental Value
Burn Time	4.277 s
Average Thrust (impulse / burn time)	≈1615 N

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=== Thrust Curve Summary ===
                        File  Peak Thrust (N)  Time of Peak (s)  Total Impulse (N·s)  Burn Time (s)
nakuja_thrust CURVE1 (2).csv       4154.46000             0.822          6909.258915          4.277
N4_4grain_simulation (1).csv       2434.85778             1.710         10041.697104          4.560
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Figure 1: Thrust curve summary output for the experimental curve (nakuja_thrust CURVE1) and the N4_4grain simulation.

5.2 Thrust Curve Comparison

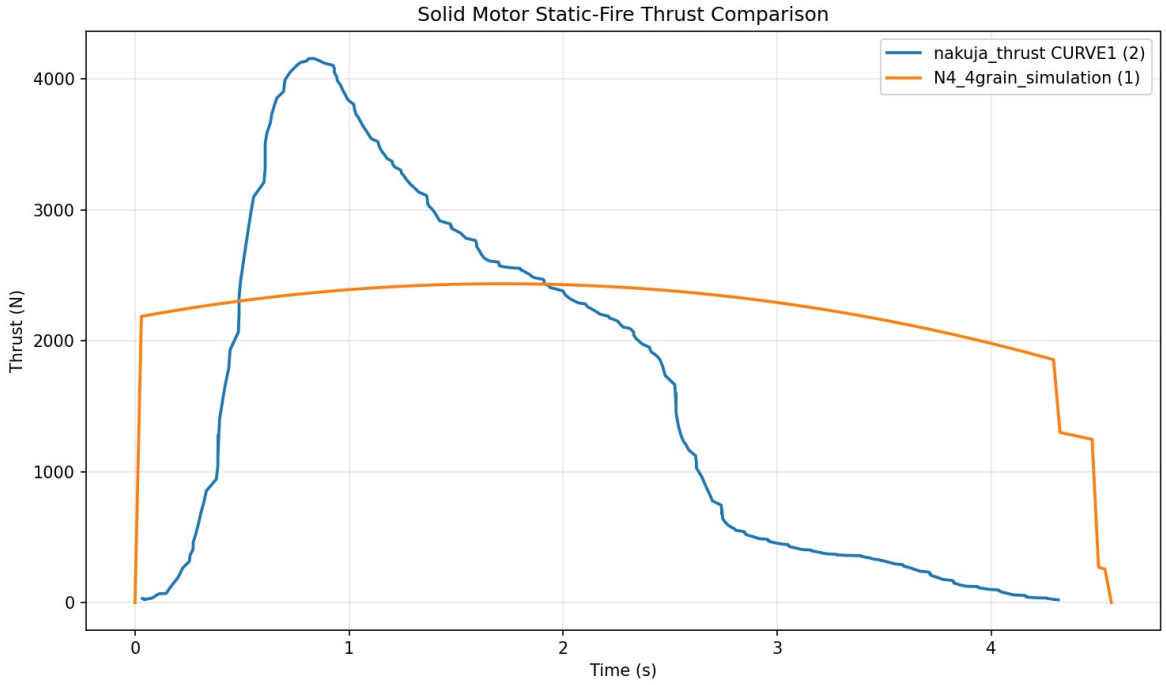


Figure 2: Solid motor static-fire thrust comparison for the N4 four-grain test (16 October 2026): experimental (blue) vs. simulated N4_4grain prediction (orange).

The experimental curve rises steeply to a sharp peak of 4154 N at 0.822 s, then declines continuously: about 2500 N by 1.9 s, about 500 N by 2.8 s, and a long low-thrust tail of a few hundred newtons out to roughly 4.3 s. The simulation instead predicts a nearly flat, slightly progressive plateau around 2200–2430 N with a late stepped tail-off. The two curves cross near 0.5 s and again near 1.9 s.

5.3 Theoretical vs. Experimental Comparison

Parameter	Simulated	Experimental	Deviation
Peak Thrust (N)	2434.858	4154.460	$+70.6\%$
Time of Peak (s)	1.710	0.822	-51.9%
Total Impulse (Ns)	10041.697	6909.259	-31.2%
Burn Time (s)	4.560	4.277	-6.2%
Average Thrust (N)	2202.1	1615.4	-26.6%
Peak-to-Average Thrust Ratio	1.11	2.57	—

5.4 Discussion

Burn time. The experimental burn time of 4.277 s is within about 6% of the simulated 4.560 s. This is a much closer match to theory than the earlier three-grain test, where the burn ran about 48% longer than predicted.

Peak thrust and curve shape. Peak thrust was about 71% above the simulated peak and occurred roughly twice as early. The peak-to-average ratio of 2.57 (against 1.11 simulated) shows a strongly regressive burn: a large share of the impulse was delivered in the first 1.5 s. Thrust is related to chamber pressure by $F = C_f P_c A_t$. With the recorded 25 mm throat ($A_t \approx 491$ mm^2), the peak thrust of 4154 N, and a thrust coefficient of about 1.5 (the ideal value for either possible nozzle expansion ratio, reduced by 5% for losses), the peak chamber pressure works out to about 5.5–5.7 MPa. No chamber pressure was measured, so this is a first-order estimate. For a thin-walled tube the hoop stress is $\sigma = PD_i/(2t)$; with a 108 mm inner diameter and 3 mm wall this gives roughly 100 MPa, and the axial force on the closures is about 50 kN. Stainless Steel 304 has a minimum yield strength of about 205 MPa (annealed), so the margin on the casing wall is roughly a factor of two before allowing for weld quality (if the tube is welded), strength loss at temperature, and local heating. The 50 kN axial load must also be carried by the bulkhead retention and acts on the sealed joint (Section 6). Peak pressure should be measured directly on the next test.

Possible role of erosive burning. The low initial port-to-throat ratio (about 1.85, Section 3.3) offers a plausible explanation for the strongly regressive shape. High gas velocity through the aft grain cores raises the burn rate early in the burn, which would produce a high, early pressure and thrust peak; as the cores widen the velocity and the burn rate fall, giving the long declining tail. A simulation using a plain burn rate law without an erosive term would not reproduce this. Early grain cracking, which increases the burning area, or pressure loss through the bulkhead leak later in the burn could also contribute. The present data cannot separate these, which is another reason to measure chamber pressure.

Impulse. Delivered impulse was about 3130 Ns (31%) below the simulation, even though burn time was close. A lower average thrust over a similar duration means less total momentum was delivered through the nozzle than predicted. The leakage at the bulkhead is a plausible contributor, since gas escaping through the joint does not pass through the nozzle and also lowers chamber pressure. Other contributors cannot be excluded from this data, including unburned or partially burned propellant, nozzle erosion, and differences between the as-built and simulated grain geometry. The effect of the leak on the measured axial force also depends on the direction of the leak jet relative to the thrust axis, which should be confirmed from the video. The specific impulse also cannot be quoted reliably until the propellant mass is confirmed (Section 3.3).

Comparison with the three-grain test.

Deviation from simulation	Three-grain (17 Aug)	Four-grain (16 Oct)
Peak thrust	-18.9%	$+70.6\%$
Time of peak	$+56.0\%$	-51.9%
Total impulse	-10.3%	-31.2%
Burn time	$+48.4\%$	-6.2%

The two tests deviate in opposite directions on peak thrust, time of peak, and burn time. The three-grain burn was slower and gentler than simulated; the four-grain burn was faster and more intense at the start. Since the grains, hardware, and leak behaviour all differ between the tests, this data alone does not isolate a single cause. Chamber pressure data is needed to separate propellant effects (burn rate, grain integrity) from hardware effects (leakage, nozzle behaviour).

Data quality. The experimental trace shows several near-vertical, stair-stepped segments (for example near 0.4 s, 0.6 s, and 2.5 s). These suggest sample-rate or resolution limits in the load cell and logging chain at high thrust and should be reviewed together with the logging configuration, in line with the team's earlier load cell sensitivity review.

6 Anomaly: Gas Leakage at the Bulkhead Side

6.1 Observation

Shortly after ignition, combustion gas was observed leaking from the bulkhead side of the motor. The burn nevertheless continued to completion. The exact onset time, duration, and leak location were not recorded with the test data and should be extracted from the video footage.

6.2 Suspected Causes

The team's working hypothesis is a combination of two mechanisms:

1. **Differential thermal expansion between the bulkhead and the nozzle.** The joint now involves a Stainless Steel 304 casing and mild steel end hardware, which have different expansion coefficients (Section 3.4). As the assembly heats, the clearances at the sealed joint change, which can reduce the squeeze on the O-ring.
2. **O-ring failure.** A seal that loses squeeze, is heated beyond its temperature rating, or is displaced under pressure will no longer contain the combustion gas, which then finds a leak path through the joint.

Illustration of the expansion mismatch. Using the typical coefficients above, a joint of 108 mm diameter that rises 100 K in temperature grows by about $17.3 \times 10^{-6} \times 108$ mm $\times$ 100 K ≈ 0.19 mm in stainless steel but only about $12 \times 10^{-6} \times 108$ mm $\times$ 100 K ≈ 0.13 mm in mild steel. The difference is roughly 0.06 mm per 100 K, in the direction that opens the radial gap between a stainless casing and a mild steel

bulkhead. The real temperature rise differs between parts (mild steel conducts heat about three times faster than stainless), so this is an order-of-magnitude illustration, not a prediction. Even so, a gap change of a few hundredths to a few tenths of a millimetre is a meaningful fraction of the squeeze on a small-section O-ring.

Alternative or contributing causes to rule out. Because the leak appeared shortly after ignition, before significant heat soak of the hardware, other mechanisms should be checked as well:

- Pressure-driven extrusion of the O-ring into the joint clearance during the high-pressure early phase (the thrust peak occurred at 0.822 s and was about 71% above the simulated peak).
- Elastic deflection of the casing or bulkhead under pressure, opening the clearance.
- O-ring material not rated for the local temperature, or direct flame or hot-gas impingement on the O-ring.
- Groove dimensions, surface finish, or O-ring size giving insufficient initial squeeze; damage to the O-ring during assembly; or misalignment of the bulkhead.

6.3 Evidence to Collect

To confirm the cause, the following should be recorded during post-test inspection:

- Condition of each O-ring (melted, hardened, cut, extruded, or flame-damaged) and which side of the ring shows the damage.
- Condition of the O-ring groove and mating surfaces: erosion tracks, soot streaks showing the leak path, discolouration, and any distortion.
- Measured groove and mating dimensions on the as-used parts, compared with the design values, including confirmation of the bulkhead outer diameter (recorded as 110 mm against a 108 mm casing bore).
- Leak onset time and location from the video, synchronised with the thrust data.

7 Outcome Classification

This test is classified as a **SUCCESS** with one noted anomaly. The pure KNSB four-grain motor ignited and burned to completion with a burn time of 4.277 s (within about 6% of simulation), and the revised stainless steel and mild steel hardware survived the burn, whereas the aluminium bulkhead from the previous four-grain test had melted. The gas leakage at the bulkhead side shortly after ignition is recorded as an anomaly that must be corrected before the next test. Impulse was about 31% below simulation and peak thrust about 71% above, and the cause of this deviation cannot yet be separated from the effect of the leak.

8 Recommendations and Corrective Actions

8.1 Corrective Action 1: Add an Additional O-Ring Layer

Add a second O-ring in series with the existing seal at the bulkhead joint, so that the joint has a redundant barrier. If the primary ring loses squeeze or fails, the secondary ring remains in place to contain the gas.

- Machine a separate second groove rather than stacking two rings in one groove, with enough spacing from the first groove to keep the wall between them structurally sound.
- Confirm the groove dimensions give an adequate static squeeze on the ring (typically in the range of roughly 15–30% for static seals, to be checked against the O-ring supplier's data), and that the groove is not overfilled when the ring swells or heats.
- Verify the O-ring material and its temperature rating, and select a high-temperature elastomer rated well above the expected local joint temperature. Previous burns have shown casing wall temperatures in the 200 °C range, and hardware close to the combustion chamber can run hotter.
- Consider an anti-extrusion backup ring on the low-pressure side of the grooves if inspection shows the primary O-ring was extruded into the clearance.
- Use new O-rings for every firing; do not re-use rings that have been heated or compressed in a previous burn.
- Apply a thin film of a compatible high-temperature lubricant during assembly, and inspect each ring for cuts or flats before installation.

8.2 Corrective Action 2: Apply a Fire-Resistant Sealant

Apply a fire-resistant sealant at the bulkhead–casing and bulkhead–nozzle joints as a supplementary barrier. The sealant fills small clearances and surface imperfections and slows or blocks hot gas if the O-rings are compromised.

- Choose a sealant rated for continuous service above the expected local joint temperature with margin, and compatible with stainless steel, mild steel, and the chosen O-ring material.
- Treat the sealant as a supplement to the O-rings, not a replacement: it should not be the only seal.
- Apply a thin, continuous bead per the manufacturer's instructions, on clean, dry, degreased surfaces, and allow the full cure time before the test.
- Keep sealant out of the chamber, the nozzle throat, and the O-ring grooves so that it does not obstruct the flow path or prevent the O-rings from seating.
- Choose a product that still allows the joint to be disassembled, and record the product name, batch, application date, and cure time in the motor build record.

8.3 Verification Before the Next Test

- Perform a cold leak check on the assembled joint (pressurised air at a pressure agreed with the Safety Officer, with leak-detection solution on the joints and the pressure held for a set time) after both corrective actions are fitted.
- Re-check the squeeze and clearance at the joint by measurement after assembly, not only by design values.
- Fit a thermocouple on the bulkhead and nozzle joint area to measure actual joint temperatures during the burn, which will confirm or rule out the differential expansion hypothesis.

8.4 Instrumentation and Data Recommendations

- Instrument chamber pressure on the next four-grain test. Peak thrust was about 71% above simulation, so pressure data is needed to check the structural margin of the casing and end hardware and to separate propellant effects from leakage effects.
- Review the load cell logging rate and resolution given the stair-stepped segments in the thrust trace.
- Synchronise the video with the thrust data so the leak onset can be placed on the thrust curve.

8.5 Records to Complete for This Test

The following items were not included with the data supplied for this report and should be added to the motor build record:

- The individual weighed mass of each of the four grains and the total propellant mass (reported as about 1 kg each; the grain geometry suggests about 2 kg each).
- Igniter mass.
- Nozzle exit diameter, and confirmation of whether the 30° and 15° nozzle angles are half-angles or included angles.
- Confirmation of the bulkhead outer diameter against the casing inner diameter, and the groove and clearance dimensions at the sealed joint.
- O-ring material, size, number, and groove dimensions as used in this test.
- Once grain masses are recorded, derive delivered specific impulse and check it against the physically plausible range for KNSB.
